

Minimal Main Body for Profile Sieve EL

Minimum Geometry Needed to Understand the Proof

Standalone main-body version

Contents

1	Model and Geometry	2
2	Paper-Facing Algorithm	3
3	Three Bridge Lemmas	4
4	The Good Event	6
5	Deterministic Geometry on the Good Event	7
6	Main Theorem	8
7	Efficiency Add-On	9
8	What This Minimal Body Leaves to the Full Version	10

1 Model and Geometry

The finite sample is viewed as the empirical Hilbert space

$$L^2(P_T) = (\mathbb{R}^T, \langle a, b \rangle_T = T^{-1}a'b), \quad \|a\|_T^2 = \langle a, a \rangle_T.$$

The population limit is

$$L^2(P), \quad \langle f, g \rangle_P = \mathbb{E}\{fg\}.$$

The sample model is

$$\mathbf{y} = \beta_0 \mathbf{x} + \boldsymbol{\mu}_0 + \mathbf{u},$$

where

$$\mu_{0,t} = m_{j_0(t)}(W_{t-1}), \quad j_0(t) = j \iff k_{j,0} < t \leq k_{j+1,0}.$$

Thus $\boldsymbol{\mu}_0$ is the stacked nuisance mean. If there are no nuisance breaks, $\mu_{0,t} = m(W_{t-1})$. With nuisance breaks, the function changes by regime.

A candidate partition is

$$\Lambda = (k_1, \dots, k_q), \quad 0 = k_0 < k_1 < \dots < k_q < k_{q+1} = T,$$

with segments

$$I_j(\Lambda) = \{t : k_j < t \leq k_{j+1}\}.$$

Let P be the $T \times K$ sieve matrix with row

$$P_K(W_{t-1})' = (p_1(W_{t-1}), \dots, p_K(W_{t-1})).$$

Let $D_j(\Lambda)$ be the diagonal selector for segment $I_j(\Lambda)$, and define the block-sieve design

$$\mathbf{Q}_{K,\Lambda} = [D_0(\Lambda)P, \dots, D_q(\Lambda)P].$$

The candidate nuisance subspace is

$$\mathcal{N}_{K,\Lambda,T} = \text{col}(\mathbf{Q}_{K,\Lambda}) \subset L^2(P_T).$$

Let $\Pi_{K,\Lambda,T}$ be the empirical orthogonal projection onto this subspace and

$$\mathbf{M}_{K,\Lambda} = I - \Pi_{K,\Lambda,T}$$

the residual-maker. The proof is governed by the subspace-valued map

$$\Lambda \longmapsto \mathcal{N}_{K,\Lambda,T}.$$

For a trial β , define

$$\mathbf{y}_\beta = \mathbf{y} - \beta \mathbf{x}.$$

The profiled residual $\mathbf{M}_{K,\Lambda} \mathbf{y}_\beta$ and the residualized score direction $\mathbf{M}_{K,\Lambda} \mathbf{w}$ both lie in the normal space

$$\mathcal{N}_{K,\Lambda,T}^\perp.$$

Inference for β is therefore carried by the projected moment

$$\Psi_T(\beta, \Lambda) = \langle \mathbf{M}_{K,\Lambda}(\mathbf{y} - \beta \mathbf{x}), \mathbf{M}_{K,\Lambda} \mathbf{w} \rangle_T.$$

2 Paper-Facing Algorithm

The paper-facing procedure is null-imposed. For a tested value β , all partition comparisons are computed using

$$\mathbf{y}_\beta = \mathbf{y} - \beta \mathbf{x}.$$

For a p -value, $\beta = \beta_0$. For a confidence set, the same profiling step is repeated at each candidate β .

The main theorem searches only $q \in \{0, 1\}$: no nuisance break versus one nuisance break. Multiple-break searches are exploratory unless the additional multi-partition localization and order-selection conditions are proved. The score weight has the generic form

$$w_t = \omega(Z_t),$$

where ω is any member of the admissible score-weight class satisfying the boundedness or max-score condition, projected variance nondegeneracy, and the projected LLN/CLT assumptions.

For each $q \in \{0, 1\}$, select

$$\hat{\Lambda}_q(\beta) \in \arg \min_{\Lambda \in \mathcal{L}_q} RSS_K(\beta, \Lambda),$$

where

$$RSS_K(\beta, \Lambda) = T \|\mathbf{M}_{K, \Lambda}(\mathbf{y} - \beta \mathbf{x})\|_T^2.$$

Then choose the order by the workbook penalty

$$\hat{q}(\beta) \in \arg \min_{q \in \{0, 1\}} \left[RSS_K\{\beta, \hat{\Lambda}_q(\beta)\} + q \kappa_T R_T \right], \quad \hat{\Lambda}(\beta) = \hat{\Lambda}_{\hat{q}(\beta)}(\beta).$$

The scalar EL scores are

$$g_t(\beta, \hat{\Lambda}) = \{\mathbf{M}_{K, \hat{\Lambda}(\beta)}(\mathbf{y} - \beta \mathbf{x})\}_t \{\mathbf{M}_{K, \hat{\Lambda}(\beta)} \mathbf{w}\}_t.$$

The EL multiplier solves

$$\sum_{t=1}^T \frac{g_t(\beta, \hat{\Lambda})}{1 + \lambda(\beta) g_t(\beta, \hat{\Lambda})} = 0,$$

and the statistic is

$$\ell_T(\beta) = 2 \sum_{t=1}^T \log\{1 + \lambda(\beta) g_t(\beta, \hat{\Lambda})\}.$$

Assumption 1 (Primitive nuisance-covariate conditions). *The nuisance covariate and sieve are required to satisfy the primitive conditions that certify the good-event objects used below.*

1. *The lagged design is predetermined:*

$$X_{t-1}, W_{t-1} \in \mathcal{F}_{t-1}, \quad E(U_t \mid \mathcal{F}_{t-1}) = 0.$$

Here W_{t-1} is the nuisance covariate; the score weight w_t is a separate admissible direction.

2. W_t has a stable marginal law P_W and satisfies the mixing or uniform empirical-law conditions needed for the searched block-sieve and break classes.
3. The basis $P_K(W) = (p_1(W), \dots, p_K(W))'$ includes an intercept, has envelope $\zeta_K = \sup_w \|P_K(w)\|$, and yields uniformly nonsingular segment Gram matrices over admissible partitions.
4. Each regime nuisance m_j is approximable by the sieve: for some $a_K \downarrow 0$,

$$\max_j \inf_{c_j \in \mathbb{R}^K} \|m_j(W) - P_K(W)'c_j\|_{L^2(P_W)} \leq a_K.$$

The resulting true-partition remainder is negligible in norm, score, and variance at the rates required by the good event.

5. The paper-facing one-break case has minimum spacing and detectable break strength

$$\min(k_0, T - k_0) \geq \epsilon T, \quad \Delta_T^2 = \|m_1(W) - m_0(W)\|_{L^2(P_W)}^2,$$

with $R_T = K + \log T + Ta_K^2$, localization radius r_T , and penalty $\kappa_T R_T$ satisfying the rate restrictions used for RSS localization, estimated-partition replacement, and order selection.

3 Three Bridge Lemmas

The following three finite-sample facts are the minimum material needed before the good-event proof.

Lemma 1 (Profiling is distance to a nuisance subspace). *For fixed (β, Λ) ,*

$$RSS_K(\beta, \Lambda) = T \text{dist}_T^2(\mathbf{y} - \beta \mathbf{x}, \mathcal{N}_{K, \Lambda, T}) = T \|\mathbf{M}_{K, \Lambda}(\mathbf{y} - \beta \mathbf{x})\|_T^2.$$

Proof. The profile RSS minimizes over a separate sieve coefficient vector c_j on each segment:

$$RSS_K(\beta, \Lambda) = \min_{c_0, \dots, c_q} \sum_{j=0}^q \sum_{t \in I_j(\Lambda)} \{Y_t - \beta X_{t-1} - P_K(W_{t-1})'c_j\}^2.$$

Stacking $c = (c'_0, \dots, c'_q)'$, the fitted nuisance vector is $\mathbf{Q}_{K, \Lambda} c$. Hence

$$RSS_K(\beta, \Lambda) = \min_c \|\mathbf{y} - \beta \mathbf{x} - \mathbf{Q}_{K, \Lambda} c\|_2^2 = \inf_{g \in \mathcal{N}_{K, \Lambda, T}} \|\mathbf{y}_\beta - g\|_2^2.$$

Since $\|a\|_2^2 = T\|a\|_T^2$, and since the orthogonal projection gives the nearest point in $\mathcal{N}_{K, \Lambda, T}$, the claim follows. $\square$

Lemma 2 (Projected moment and population identification). *Let the true population nuisance space be*

$$\mathcal{H}_0 = \left\{ h(A) = \sum_{j=0}^{q_0} h_j(W) \mathbf{1}\{\tau_{j,0} < S \leq \tau_{j+1,0}\} : h_j \in L^2(P_W) \right\},$$

where $A = (S, W)$. Let M_0^P be the population residual-maker onto $\mathcal{H}_0^\perp$. If $\mu_0 \in \mathcal{H}_0$, $\mathbb{E}(U \mid X, A) = 0$, and

$$\Gamma = \langle M_0^P X, M_0^P w \rangle_P \neq 0,$$

then

$$\Psi(\beta, \Lambda_0) = \langle M_0^P (Y - \beta X), M_0^P w \rangle_P = -(\beta - \beta_0)\Gamma.$$

Therefore β_0 is the unique population zero.

Proof. At the truth,

$$Y - \beta X = \mu_0 + U - (\beta - \beta_0)X.$$

Because $\mu_0 \in \mathcal{H}_0$, $M_0^P \mu_0 = 0$. Because $M_0^P w$ is a function orthogonal to the nuisance space and $\mathbb{E}(U \mid X, A) = 0$, the population exogeneity condition gives

$$\langle M_0^P U, M_0^P w \rangle_P = 0.$$

Thus

$$\Psi(\beta, \Lambda_0) = -(\beta - \beta_0)\langle M_0^P X, M_0^P w \rangle_P = -(\beta - \beta_0)\Gamma.$$

If $\Gamma \neq 0$, this equals zero only when $\beta = \beta_0$. □

Lemma 3 (Scalar EL is convex KKT). *For fixed (β, Λ) , define*

$$g_t(\beta, \Lambda) = \{\mathbf{M}_{K,\Lambda}(\mathbf{y} - \beta \mathbf{x})\}_t \{\mathbf{M}_{K,\Lambda} \mathbf{w}\}_t.$$

The one-dimensional empirical likelihood problem is

$$\max_{\pi \in \Delta_T} \sum_{t=1}^T \log(T\pi_t) \quad \text{subject to} \quad \sum_{t=1}^T \pi_t g_t(\beta, \Lambda) = 0.$$

If zero lies in the interior of the convex hull of $\{g_t\}_{t=1}^T$, the KKT weights have the form

$$\pi_t(\lambda) = \frac{1}{T\{1 + \lambda g_t\}}, \quad P_T \left\{ \frac{g_t}{1 + \lambda g_t} \right\} = 0,$$

where $P_T a_t = T^{-1} \sum_t a_t$. If additionally

$$P_T g_t = O(T^{-1/2}), \quad P_T g_t^2 \geq c > 0, \quad \max_t |g_t|/\sqrt{T} = o(1),$$

then

$$-2 \log EL(\beta, \Lambda) = \frac{T(P_T g_t)^2}{P_T g_t^2} + o(1).$$

Proof. The feasible set is the intersection of the probability simplex with the affine moment hyperplane $\sum_t \pi_t g_t = 0$. The Lagrangian first-order condition for π_t , together with $\sum_t \pi_t = 1$, gives

$$\pi_t = \frac{1}{T(1 + \lambda g_t)}$$

for a scalar multiplier λ . The moment restriction gives the displayed dual equation. Under the size conditions, the dual equation implies

$$\lambda = \frac{P_T g_t}{P_T g_t^2} + o(T^{-1/2})$$

and $\max_t |\lambda g_t| = o(1)$. Expanding

$$-2 \log EL(\beta, \Lambda) = 2 \sum_{t=1}^T \log(1 + \lambda g_t)$$

to second order gives the quadratic formula. □

4 The Good Event

The proof separates deterministic geometry from probability. Let $\delta_T \downarrow 0$. The good event is

$$\mathcal{G}_T = G_T^{\text{iso}} \cap G_T^{\text{approx}} \cap G_T^{\text{angle}} \cap G_T^{\text{local}} \cap G_T^{\text{select}} \cap G_T^{\text{convex}}.$$

The components mean:

1. G_T^{iso} : empirical Gram matrices and the empirical inner products used in the score, derivative, and variance are stable on the searched subspaces.
2. G_T^{approx} : the true nuisance is close to the true break-aware sieve subspace. With $\mathbf{M}_0 = \mathbf{M}_{K, \Lambda_0}$, $\mathbf{r}_{\mu, T} = \mathbf{M}_0 \boldsymbol{\mu}_0$ satisfies

$$\|\mathbf{r}_{\mu, T}\|_T \leq \delta_T, \quad T^{-1/2} |\mathbf{r}'_{\mu, T} \mathbf{M}_0 \mathbf{w}| \leq \delta_T.$$

3. G_T^{angle} : the population transversality constant $|\Gamma|$ is bounded away from zero, and the oracle score variance is nondegenerate. With $\mathbf{v}_0 = \mathbf{M}_0 \mathbf{w}$,

$$A_T^0 = T^{-1/2} \mathbf{u}' \mathbf{v}_0, \quad B_T^0 = T^{-1} \sum_{t=1}^T u_t^2 v_{0,t}^2,$$

we have $|A_T^0| \leq C$ and $B_T^0 \geq c > 0$.

4. G_T^{local} : the estimated partition can replace the true partition in the only directions needed by the score:

$$\begin{aligned} \sqrt{T} |\Psi_T(\beta_0, \hat{\Lambda}) - \Psi_T(\beta_0, \Lambda_0)| &\leq \delta_T, \\ |V_T(\hat{\Lambda}) - V_T(\Lambda_0)| &\leq \delta_T, \quad \sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \hat{\Lambda}) - \Psi_T(\beta, \Lambda_0)| \leq \delta_T. \end{aligned}$$

Equivalently, residual-makers are close in the directional topology actually used by the proof, not necessarily in global operator norm.

5. G_T^{select} : RSS selects a local break partition and, under unknown order, the penalized criterion selects q_0 .

6. G_T^{convex} : the scalar EL moment cloud straddles zero and $\max_t |g_t(\beta_0, \hat{\Lambda})|/\sqrt{T} \leq \delta_T$, so the EL KKT root exists and the logarithmic barrier expansion is valid.

Component	Used for
G_T^{iso}	stable projection geometry and denominator control
G_T^{approx}	remove the nuisance from the oracle score
G_T^{angle}	identification, nonzero derivative, nonzero variance
G_T^{local}	replace Λ_0 by $\hat{\Lambda}$
G_T^{select}	justify break localization and order selection
G_T^{convex}	justify EL KKT existence and quadratic expansion

The full-detail version proves

$$\mathbb{P}(\mathcal{G}_T) \rightarrow 1.$$

The main body uses only the deterministic implications of $\mathcal{G}_T$.

5 Deterministic Geometry on the Good Event

Theorem 1 (Consistency on $\mathcal{G}_T$). *Suppose Lemma 2 holds. On $\mathcal{G}_T$, if*

$$|\Psi_T(\hat{\beta}, \hat{\Lambda})| \leq \delta_T,$$

then

$$|\hat{\beta} - \beta_0| \leq \frac{2\delta_T}{|\Gamma|}.$$

Consequently, if $\mathbb{P}(\mathcal{G}_T) \rightarrow 1$, then $\hat{\beta} \rightarrow_p \beta_0$.

Proof. On $\mathcal{G}_T$, the uniform bridge between sample and population moments gives

$$\sup_{\beta \in \mathcal{B}} |\Psi_T(\beta, \hat{\Lambda}) - \Psi(\beta, \Lambda_0)| \leq \delta_T.$$

Therefore

$$|\Psi(\hat{\beta}, \Lambda_0)| \leq |\Psi(\hat{\beta}, \Lambda_0) - \Psi_T(\hat{\beta}, \hat{\Lambda})| + |\Psi_T(\hat{\beta}, \hat{\Lambda})| \leq 2\delta_T.$$

By Lemma 2,

$$|\Psi(\hat{\beta}, \Lambda_0)| = |\hat{\beta} - \beta_0||\Gamma|,$$

so the claim follows. □

Theorem 2 (Wilks reduction on $\mathcal{G}_T$). *Let*

$$V_T(\Lambda) = P_T g_t(\beta_0, \Lambda)^2.$$

On $\mathcal{G}_T$,

$$-2 \log EL(\beta_0, \hat{\Lambda}) = \frac{(A_T^0)^2}{B_T^0} + o(1),$$

where

$$A_T^0 = T^{-1/2} \mathbf{u}' \mathbf{v}_0, \quad B_T^0 = T^{-1} \sum_{t=1}^T u_t^2 v_{0,t}^2, \quad \mathbf{v}_0 = \mathbf{M}_0 \mathbf{w}.$$

If additionally

$$A_T^0 \Rightarrow N(0, V), \quad B_T^0 \rightarrow_p V > 0,$$

then

$$-2 \log EL(\beta_0, \hat{\Lambda}) \Rightarrow \chi_1^2.$$

Proof. By Lemma 3 and G_T^{convex} ,

$$-2 \log EL(\beta_0, \hat{\Lambda}) = \frac{T \{\Psi_T(\beta_0, \hat{\Lambda})\}^2}{V_T(\hat{\Lambda})} + o(1).$$

By G_T^{local} , $\hat{\Lambda}$ can be replaced by Λ_0 in both numerator and denominator. At the true partition,

$$\sqrt{T} \Psi_T(\beta_0, \Lambda_0) = T^{-1/2} \mathbf{u}' \mathbf{v}_0 + T^{-1/2} \mathbf{r}'_{\mu, T} \mathbf{v}_0 = A_T^0 + o(1),$$

where the last equality uses G_T^{approx} . Also

$$V_T(\Lambda_0) = B_T^0 + o(1)$$

by G_T^{iso} and the variance part of G_T^{angle} . Substitution gives the good-event reduction. The chi-square limit follows by Slutsky. $\square$

Proposition 1 (Break search is subspace selection). *By Lemma 1, the break estimator chooses the nuisance subspace closest to $\mathbf{y} - \beta \mathbf{x}$:*

$$\hat{\Lambda}_q(\beta) \in \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} \text{dist}_T^2(\mathbf{y} - \beta \mathbf{x}, \mathcal{N}_{K, \Lambda, T}).$$

On G_T^{select} , this nearest-subspace selector is local around Λ_0 , and the penalized selector chooses q_0 .

6 Main Theorem

Theorem 3 (Profile-sieve EL main theorem). *Assume:*

1. the model and population identification conditions in Lemma 2;
2. the stochastic regularity conditions imply $\mathbb{P}(\mathcal{G}_T) \rightarrow 1$;
3. the oracle projected score satisfies

$$A_T^0 = T^{-1/2} \mathbf{u}' \mathbf{M}_0 \mathbf{w} \Rightarrow N(0, V), \quad B_T^0 = T^{-1} \sum_{t=1}^T u_t^2 (\mathbf{M}_0 \mathbf{w})_t^2 \rightarrow_p V > 0;$$

4. the estimator satisfies the local projected moment condition, with probability tending to one,

$$|\Psi_T(\hat{\beta}, \hat{\Lambda})| \leq \delta_T.$$

Then

$$\hat{\beta} \rightarrow_p \beta_0$$

and

$$-2 \log EL(\beta_0, \hat{\Lambda}) \Rightarrow \chi_1^2.$$

Proof. Consistency follows from Theorem 1 because $\mathbb{P}(\mathcal{G}_T) \rightarrow 1$. The Wilks result follows from Theorem 2, the oracle score CLT, and the oracle variance LLN. $\square$

7 Efficiency Add-On

Efficiency is not needed for Wilks validity. It requires an additional efficiency model. Let $A = (S, W)$, and suppose the conditional Gaussian location calculation is valid:

$$U \mid X, A \sim N(0, \sigma^2(A)), \quad 0 < c \leq \sigma^2(A) \leq C < \infty.$$

The raw score is $UX/\sigma^2(A)$, and the nuisance tangent directions are

$$Uh(A)/\sigma^2(A), \quad h \in \mathcal{H}_0.$$

Projecting the raw score away from the nuisance tangent space gives

$$S_{\text{eff}} = \frac{U\{X - E(X \mid A)\}}{\sigma^2(A)}.$$

The efficient information is

$$I_{\text{eff}} = \mathbb{E} \left[\frac{\{X - E(X \mid A)\}^2}{\sigma^2(A)} \right].$$

To make the projected sample score equal this efficient score, choose

$$w^*(X, A) = \frac{X}{\sigma^2(A)}.$$

Because $\sigma^2(A)$ is A -measurable,

$$M_0^P w^* = \frac{X - E(X \mid A)}{\sigma^2(A)}.$$

Thus, at β_0 ,

$$\{M_0^P(Y - \beta_0 X)\} \{M_0^P w^*\} = \frac{U\{X - E(X \mid A)\}}{\sigma^2(A)} = S_{\text{eff}}.$$

If the sieve and estimated-break replacements preserve this score at $T^{-1/2}$ scale, then any EL or moment root using w^* has

$$\sqrt{T}(\widehat{\beta} - \beta_0) = I_{\text{eff}}^{-1} T^{-1/2} \sum_{t=1}^T \frac{U_t \{X_t - E(X_t | A_t)\}}{\sigma^2(A_t)} + o_p(1),$$

and hence attains the semiparametric efficiency bound

$$I_{\text{eff}}^{-1} = \left\{ \mathbb{E} \left[\frac{\{X - E(X | A)\}^2}{\sigma^2(A)} \right] \right\}^{-1}.$$

In the homoskedastic case, $w^* = X/\sigma^2$, so the same root is obtained by using $w = X$.

8 What This Minimal Body Leaves to the Full Version

The minimal body does not reprove the probability bounds. The full-detail body contains the supporting proofs for:

1. empirical Gram stability and empirical-to-population projection bridges;
2. sieve approximation of the true nuisance and projected score directions;
3. RSS localization and unknown-order penalty comparisons;
4. estimated-break directional replacement;
5. oracle score CLT and variance LLN;
6. max-score and convex-hull control for EL;
7. heteroskedastic efficient-weight replacement, if the efficiency theorem is used.

The role of this minimal body is to show exactly how those stochastic facts are used once they are available.